Orbital Curbside: the one-page version

Plain-language summary of the research paper that follows · Brandon Graff · Winter Park High School · October 2026

Space is filling up with junk: dead satellites and old rocket stages that can collide and shatter into thousands of fragments. Removing even one piece today is a custom mission costing **$40–93 million**. We asked: what if one spacecraft could collect many pieces on a single trip, the way a garbage truck runs a route?

10
old rocket stages removed by one electric spacecraft on our first route

$12.2M
modelled cost per removal including launch, versus $97M doing one object per mission

2.2%
difference between our physics and the 2017 world-champion (JPL) solution

The trick: Earth does the driving. Earth is slightly wider at the equator. That bulge slowly twists every orbit around the planet, and how fast depends on the orbit's height. Turning a spacecraft's orbit with rocket fuel is brutally expensive. So our spacecraft changes its height a little, waits while Earth twists its orbit into line with the next piece of junk, then hops over. At each stop it bolts on a small rocket "deorbit kit" that drops the junk into the atmosphere, where it burns up. Then it drives on.

What we built. A route-planning program that picks which objects to visit, in what order and when, using real orbits of today's catalogue (16 October 2026). We checked our physics against the winning solution of a 2017 European Space Agency competition on this exact problem, re-checked our routes with the official orbit predictor, and built a 3D simulation that flies the route.

What we found. The cheapest ride to space, a shared "rideshare" launch, only goes to one kind of orbit. So our first route targets 10 old rocket stages owned by the US, India and France that already sit in that band. From launch to the last stop takes about 16 months and 198 kg of propellant. Spreading the big fixed costs (the spacecraft, the launch and the team) over ten objects brings the modelled cost to about **$12.2 million per object**, launch included. A spacecraft that stays up for years and receives new kits brings it to about **$5.2 million**. Reaching the famous Soviet SL-8 stages costs more ($13.0 million), because the spacecraft must first spend about a year turning its orbit. Surprisingly, the choice of engine barely changes the price. What matters is **how many stops one spacecraft makes**, and that its electric engine never needs refuelling.

Being honest. The physics idea is not new: scientists published it between 2015 and 2026. What is new here is turning it into a price list on today's real debris. Our costs are estimates, and two numbers matter most and are still uncertain: what the spacecraft and the kits would really cost. Grabbing a tumbling, unprepared rocket stage has never been done. The first attempt is planned for 2027–28. We are now interviewing engineers and satellite operators to test our assumptions.

Orbital Curbside: Pricing Multi-Target Active Debris Removal with J2-Drift Routing on the 2026 Catalogue

Brandon Graff · Winter Park High School · DECA Innovation Plan 2026–27 · October 2026 · Code and data: github.com/brandongraff-dev/deca2026

Abstract. Active debris removal (ADR) is today bought one object at a time, at public contract values of $40–93M per object. Multi-target removal that exploits Earth's oblateness (J2 nodal precession) to align orbital planes "for free" is well established in the astrodynamics literature, notably in ESA's GTOC9 competition (2017), but it has not been turned into a public, customer-facing price. We build a reproducible route optimiser (closed-form drift-time legs, impulsive and Edelbaum low-thrust transfers, four metaheuristics plus dynamic-programming timing) and validate it against the winning JPL GTOC9 solution: across 113 legs our model's total delta-v is within 2.2% of JPL's ($r = 0.99$). We then apply it to the real 16 October 2026 catalogue, including the way to orbit: a rideshare to sun-synchronous orbit followed by the servicer's own climb, or a dedicated launch. The rideshare is cheapest on every route. An electric servicer that attaches solid deorbit kits removes 10 partner-nation rocket stages from the sun-synchronous band in 490 days from launch, and 10 SL-8 stages from the 83° cluster in 595 days, of which 375 are spent changing inclination after launch. SGP4 re-propagation confirms the planned plane alignments to 0.06° (SL-8) and 0.30° (sun-synchronous) median. Under stated cost assumptions the first route costs $12.2M per removal including launch (Monte Carlo P10–P90: $11.2–15.4M), versus $97M for one-object missions in the same model, and about $5.2M for a reusable servicer restocked with kit pods until its thrusters reach their rated life. The SL-8 route costs $13.0M. Electric versus chemical propulsion changes the price by only a few percent. The economic levers are stops per servicer and reuse, while electric propulsion's advantage is operational: it never needs in-space refuelling.

1. THE PROBLEM

About 47,080 objects are regularly tracked in Earth orbit, of which about 15,900 are working spacecraft; ESA's model estimates 68,450 objects larger than 10 cm [1]. ESA's 2026 environment report states that "even if no new satellites were launched, the debris population would continue to increase because fragmentation events generate debris faster than they can naturally reenter the atmosphere" [2], the self-sustaining cascade first described by Kessler and Cour-Palais [3]. A small number of massive derelicts dominates the long-term risk: McKnight et al. found that removing the 50 statistically most concerning objects, mostly Soviet-era SL-16 and SL-8 rocket stages, would roughly halve the collision risk they pose [4].

Regulation is tightening. The US Federal Communications Commission (FCC) requires LEO satellites launched after 29 September 2024 to be disposed of "as soon as practicable, and no later than five years after the end of the mission" [5]. Its July 2026 Part 100 order keeps the five-year rule and requires a disposal success probability of at least 0.9 per spacecraft [6]. In effect, up to one in ten satellites may fail to dispose of itself and still be compliant. Those failed satellites, plus the legacy rocket bodies no one owns any more, are the potential market for removal.

Removal is now being bought, but one object at a time: ESA's ClearSpace-1 (€86M, one object), JAXA/Astroscale ADRAS-J2 (about ¥13.2B, one rocket stage) and the UK's 2026 award to ClearSpace (about £61M for two satellites) [7–9]. NASA's 2023 cost–benefit study concluded that removal is rarely cost-effective at these prices, and noted that "disposable modules deployed from a mothership may provide the lowest cost option", an architecture it did not analyse [10].

2. PRIOR WORK, AND WHAT IS NEW HERE

The physics is not new. ESA's 9th Global Trajectory Optimisation Competition (GTOC9, "The Kessler Run", 2017) asked teams to remove 123 sun-synchronous debris objects with multi-rendezvous missions that deliver 30 kg deorbit packages [11]. The winning JPL team removed all 123 with 10 missions (731.28 M€) using branch-and-bound

searches that exploit nodal drift, beam search, ant-colony optimisation and a genetic algorithm [12]. Low-thrust multi-target removal using J2 drift was studied by Di Carlo, Romero Martin and Vasile [13], and a 2026 preprint by Iskender et al. formalises "drift-aware" tug logistics on the GTOC9 benchmark, within 1.55% of JPL's cost [14]. Target selection by collision risk is also established [15, 16].

What we add. We found no public work that (i) prices multi-target removal per object for customers, (ii) on the *current* catalogue, (iii) comparing an electric drift servicer, a chemical drift servicer and today's one-object missions in one consistent cost model, including (iv) a reusable servicer restocked by kit pods. Our contribution is this integration and business analysis, with a validated, open-source tool. It is not a new astrodynamics method.

3. METHOD

3.1 Data

Orbital elements come from CelesTrak's GP (TLE) data downloaded on 16 October 2026: active satellites, four major debris clouds, all rocket bodies and the legacy constellations of three operators. Metadata comes from CelesTrak's SATCAT, and masses from McDowell's GCAT [17, 18]. Elements are decoded with the official SGP4 implementation [19]. We identified "truck-friendly" clusters: many heavy objects with similar altitude and inclination but planes spread around the globe (Fig. 1, Table 1). Our counts match McKnight et al.'s published cluster sizes (for example, about 150 SL-8 stages near 950 km and 83° [4]), which cross-checks the data.

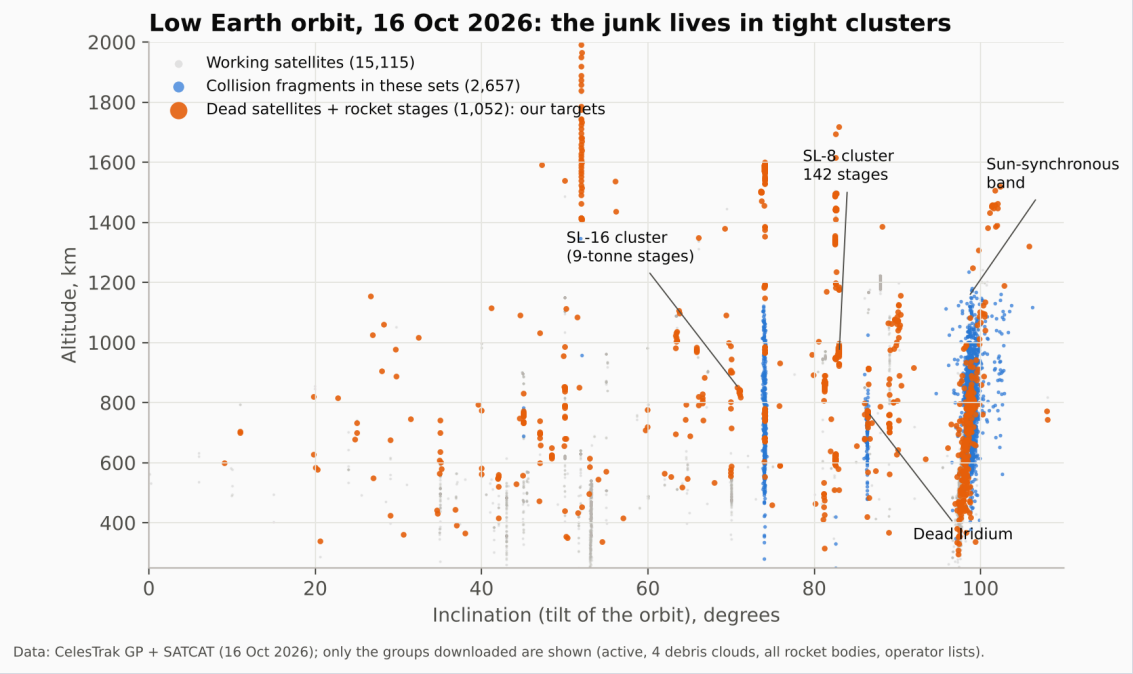


Fig. 1. Objects in low Earth orbit on 16 Oct 2026 (subset downloaded). Derelicts (orange) sit in tight clusters of inclination and altitude.

Cluster (16 Oct 2026)	Objects	Median kg	Total t	Drift °/day	Planes span
SL-16 (Zenit-2 upper stage) cluster	18	9,000	162.0	-2.12 to -2.10	317°
SL-8 (Kosmos-3M stage) 83 deg cluster	142	1,434	203.6	-0.76 to -0.74	356°
SL-8 74 deg / ~750 km cluster	50	1,434	71.7	-1.87 to -1.83	356°
SL-8 74 deg / ~1500 km cluster	45	1,434	64.5	-1.34 to -1.25	355°
Dead first-generation Iridium	24	662	15.9	-0.47 to -0.42	358°
Rocket bodies, sun-synchronous band	173	1,000	294.9	+0.74 to +1.19	359°

Table 1. Candidate clusters. Drift rates from the SGP4-consistent model.

3.2 The conveyor belt: J2 nodal precession

Earth's oblateness makes the right ascension of the ascending node Ω (the direction an orbital plane faces) drift secularly [20]:

$$\dot{\Omega} = -\frac{3}{2}\, n\, J_2 \left(\frac{R_E}{p}\right)^2 \cos i, \qquad n = \sqrt{\mu/a^3}, \quad p = a(1-e^2)$$

The rate depends on altitude and inclination (Fig. 2): $d\dot{\Omega}/da = -3.5\,\dot{\Omega}/a$ and $d\dot{\Omega}/di = -\dot{\Omega}\tan i$. A sun-synchronous orbit at 800 km comes out at +0.985°/day, matching its definition. For the real catalogue we add a small correction table built from SGP4's own secular rates (second-order J_2^2 and J_4 terms). Without it, plane predictions drift by about 1° over 1,000 days (Section 4.3).

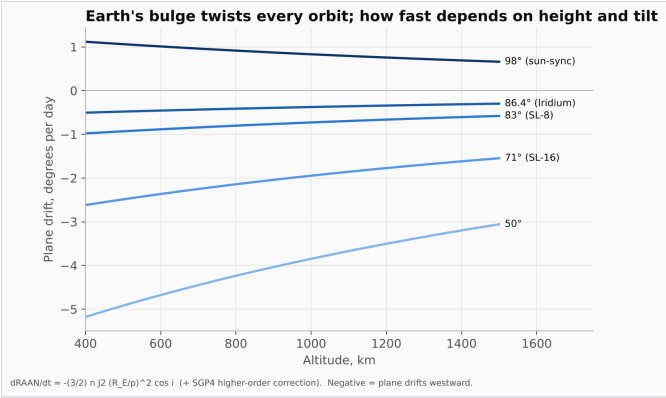


Fig. 2. Plane drift rate versus altitude for the inclinations of our target clusters.

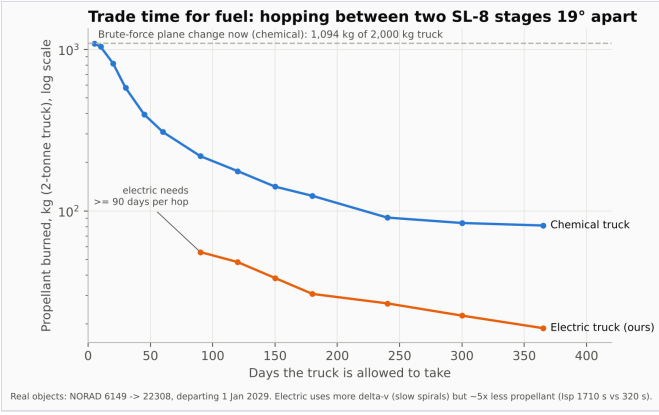


Fig. 3. Trading time for propellant on a real 19° hop between two SL-8 stages. The electric truck uses more Δv (slow spirals) but about 5× less propellant mass, and needs at least about 90 days per hop.

3.3 Transfer costs

The angle θ between two orbital planes follows from spherical trigonometry, $\cos\theta = \cos i_1 \cos i_2 + \sin i_1 \sin i_2 \cos\Delta\Omega$. We use a common form for both engine types:

$$\text{chemical (impulsive):} \quad \Delta v = \sqrt{v_1^2 + v_2^2 - 2\, v_1\, v_2 \cos\theta}$$
$$\text{electric (Edelbaum [21]):} \quad \Delta v = \sqrt{v_1^2 + v_2^2 - 2\, v_1\, v_2 \cos(\pi\theta/2)}$$

v_1, v_2 are circular speeds before and after. With θ = 0 both reduce to a small altitude change (52.45 m/s for 700 → 800 km, identical to a Hohmann transfer). A 10° plane change at 800 km costs 1.30 km/s, which is why waiting for drift is so valuable. Propellant follows the rocket equation, $m_f = m_0 \exp(-\Delta v / I_{sp} g_0)$. Electric spiral times use constant thrust: $t = (m_0 I_{sp} g_0 / F)(1 - e^{-\Delta v / I_{sp} g_0})$.

3.4 One leg: wait versus burn

Each leg from object A to object B in time T has four phases: spiral out to a drift orbit (a_d, i_d) → drift for time τ → spiral into B's orbit → ride along with B. The drift time is solved in closed form so the plane gap closes exactly:

$$G(\tau) = C + \tau\,(\dot{\Omega}_B - \dot{\Omega}_d) = 0 \pmod{360^\circ}, \qquad C = \Delta\Omega_0 + t_1(\dot{\Omega}_B - (\dot{\Omega}_A + \dot{\Omega}_d)/2) + t_2(\dot{\Omega}_B - \dot{\Omega}_d)/2$$

This generalises the classic waiting time $\Delta t = \Delta\Omega / |\dot{\Omega}_1 - \dot{\Omega}_2|$ to include spiral times t_1, t_2. If the gap cannot close within T, the remaining angle is removed by a plane change. Drift orbits are searched on a grid of altitude (600–1,500 km, avoiding the crowded shells below 600 km) and inclination (±3°). A fixed 20 m/s per stop covers final phasing and proximity operations. Fig. 3 shows the resulting trade between time and propellant for one real leg.

3.5 Which objects, in which order

Choosing and ordering 10 of the 142 SL-8 stages has about 2.4×10^{21} possibilities; ordering 10 fixed objects alone has 3.6 million. Every option changes as planes drift. We implemented beam search, simulated annealing (including moves that swap in unused objects), a genetic algorithm with order crossover, and ant-colony optimisation. All share one cost oracle. A dynamic-programming timer then chooses every leg's duration jointly, so a slightly more expensive hop can be accepted if it sets up cheaper hops later. The planner minimises $\Delta v + \lambda \cdot$(mission days). λ, the "price of time", is derived from the cost model: an extra day costs one day of operations, and an extra m/s costs its propellant plus the launch of that propellant. This gives $\lambda \approx 11$ m/s per day for the electric truck and ≈ 2.7 for the chemical truck.

3.6 Deorbit kits and the servicer

At each stop the truck attaches a kit, a small solid motor that lowers the object's perigee to 250 km (re-entry within months). Objects heavier than 5 t would instead get a controlled re-entry to 50 km. Kit mass follows from the rocket equation (I_{sp} = 285 s, 12 kg fixed plus 15% tankage, `assumption`): about 130 kg for a 1.4 t SL-8 stage, about 53 kg for a dead Iridium satellite, and about 870 kg for a 9 t SL-16. That is why SL-16 stages are not suited to a multi-stop truck. The electric servicer uses two Busek BHT-1500-class Hall thrusters (0.202 N total, I_{sp} 1,710 s) [22] and a 600 kg dry mass `assumption`, and spends 10 days at each stop `assumption`. The chemical comparison truck is identical except for a bipropellant engine with I_{sp} 320 s.

3.7 Getting to the first stop

A launch does not deliver a servicer wherever it is needed. The cheapest option, a SpaceX Transporter-class rideshare ($7,500/kg in 2027, with a custom quote above 831 kg per port [25]), goes to sun-synchronous orbit at about 500–600 km and 97.6° [26]. J2 drift rotates planes about the pole but does not change inclination, so reaching the 83° SL-8 cluster requires the servicer to change its own inclination. We model this as one more drift-orbit leg from a 550 km sun-synchronous drop-off orbit to the first target, with every kit and all propellant aboard, priced at the same economic price of time. The drop-off plane depends on the launch's local time of ascending node, which we cannot choose, so we evaluate all 24 hours and report the median. The alternatives are a dedicated Rocket Lab Neutron (average price $50–55M; first flight now expected in 2027 [27]) or Falcon 9 ($74M [28]), which can place the servicer directly in its first target's orbit. A lighter servicer can always fly a heavier one's schedule by throttling its thrusters, and the leg model allows this.

3.8 Validation

We test the model in three ways. (1) **Physics:** we re-cost all 113 legs of JPL's winning GTOC9 route using JPL's own endpoints, dates and durations. (2) **Optimiser:** we re-solve GTOC9's Mission 10 from scratch. (3) **Real-world fidelity:** we re-propagate every planned 2026 rendezvous with SGP4, which includes drag, and fly the drift orbits through a numerical integration with J_2, J_3 and J_4 gravity.

3.9 Economics

Mission cost = truck hardware + kits + launch (launch mass × price per kg) + propellant + operations (per year) + insurance. A 10% Δv margin is applied. Cost per removal is the mission cost divided by the number of objects. Each parameter has low, base and high values (Appendix A). We report a one-at-a-time sensitivity ("tornado") analysis and a 20,000-run Monte Carlo with triangular distributions. The reuse scenario keeps one servicer for several routes: after the first launch, only kit "pods" are launched (an electric truck carries all its xenon from the start; a chemical truck also needs its propellant delivered). Pods ride later rideshares, so reuse is modelled only where rideshares go: the sun-synchronous first route. The electric servicer's routes are capped by its thrusters' rated total impulse (13 MN·s for two BHT-1500-class units [22]).

4. RESULTS

4.1 Benchmark against the GTOC9 winner

Across JPL's 113 legs our model totals 24.6 km/s against JPL's 25.1 km/s (-2.2%), with per-leg correlation r = 0.99 and 90% of legs within ±25% (Fig. 4). Rebuilding JPL's launch masses from their leg Δv reproduces their published values within 0.3 kg, which checks our mass bookkeeping. Flying the same legs with an immediate plane change instead of waiting for drift costs 2.0× more Δv. Our model is slightly optimistic, mainly because it ignores phasing and eccentricity, so we apply a 10% Δv margin in all costs.

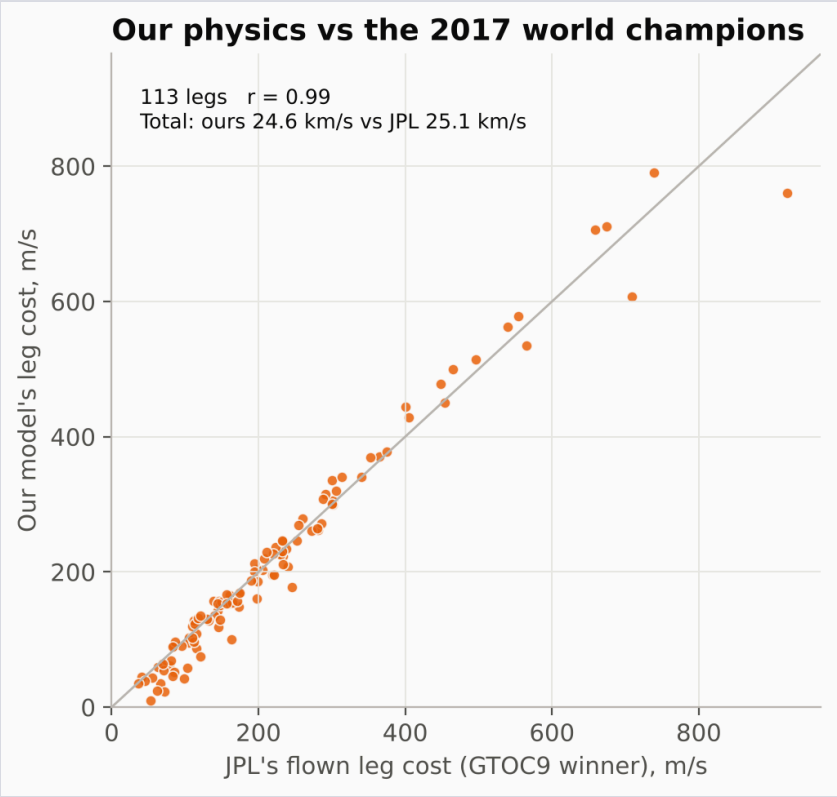


Fig. 4. Our model versus JPL's flown cost for each of the 113 legs of the winning GTOC9 solution [12].

GTOC9 Mission 10 (10 objects; all scored by our model)	Total Δv, m/s	Launch mass, kg
JPL route, our model	1,464	3,478.6
Beam search	2,012	4,040.8
Simulated annealing	1,574	3,602.5
Genetic algorithm	1,574	3,602.5
Ant colony	1,847	3,908.4
Beam + annealing polish	1,362	3,387.7
JPL order + our DP timing	1,363	3,380.5

Table 2. Optimisers on GTOC9 Mission 10. JPL's actual flown launch mass: 3,438.6 kg.

On Mission 10 (Table 2), beam search followed by simulated annealing, starting from scratch, found a route as light as JPL's own order re-timed by our dynamic-programming timer, and lighter than JPL's route evaluated in our model. Because our model is about 2% optimistic, this shows the optimiser is competitive; it does *not* show that we beat JPL. No single method dominated: beam search alone and ant colony performed worst, and the hybrid did best.

4.2 Missions on the 2026 catalogue

10-stop route, first stop 1 Jan 2029	Engine	Δv m/s	Propellant kg	Kits kg	Launch kg	Days
Partner-nation stages, sun-sync (first route)	electric	2,026	135	852	1,588	410
Partner-nation stages, sun-sync (first route)	chemical	1,717	776	840	2,216	390
SL-8 stages, 83°, ~960 km (Product B)	electric	678	56	1,327	1,982	220
SL-8 stages, 83°, ~960 km (Product B)	chemical	574	277	1,327	2,203	195
SL-8 stages, 74°, ~750 km	electric	1,020	68	979	1,648	550
SL-8 stages, 74°, ~750 km	chemical	1,028	470	978	2,048	400
Dead Iridium, 86° (Product A)	electric	3,165	179	526	1,305	555
Dead Iridium, 86° (Product A)	chemical	2,290	936	526	2,062	480

Table 3. Optimised missions on the real 2026 catalogue (simulated plans), each planned with the price of time derived from the cost model. These are the routes themselves, from the first stop; Table 5 adds the launch and the climb to the first stop.

The optimiser picks neighbouring objects whose planes are close and drift almost in step. Within the 142-object SL-8 cluster it finds ten stages whose planes lie within a few degrees of each other (Fig. 5), so each hop is cheap: the whole 10-stop route needs 678 m/s, of which 200 m/s is the fixed approach allowance. **The truck's mass is dominated by the kits, not by propellant.** For the 10-stop SL-8 route the kits weigh 1,327 kg against 56 kg of xenon. The first route draws on 26 rocket stages in the sun-synchronous band owned by the US, India, France, Japan or South Korea, of 300–2,500 kg and above 650 km (below that, drag brings stages down on its own and makes their planes hard to predict). Sun-synchronous planes all turn at almost the same rate, so the truck drifts by tilting its own orbit a few degrees; the route needs 2,026 m/s over 410 days, with 852 kg of kits.

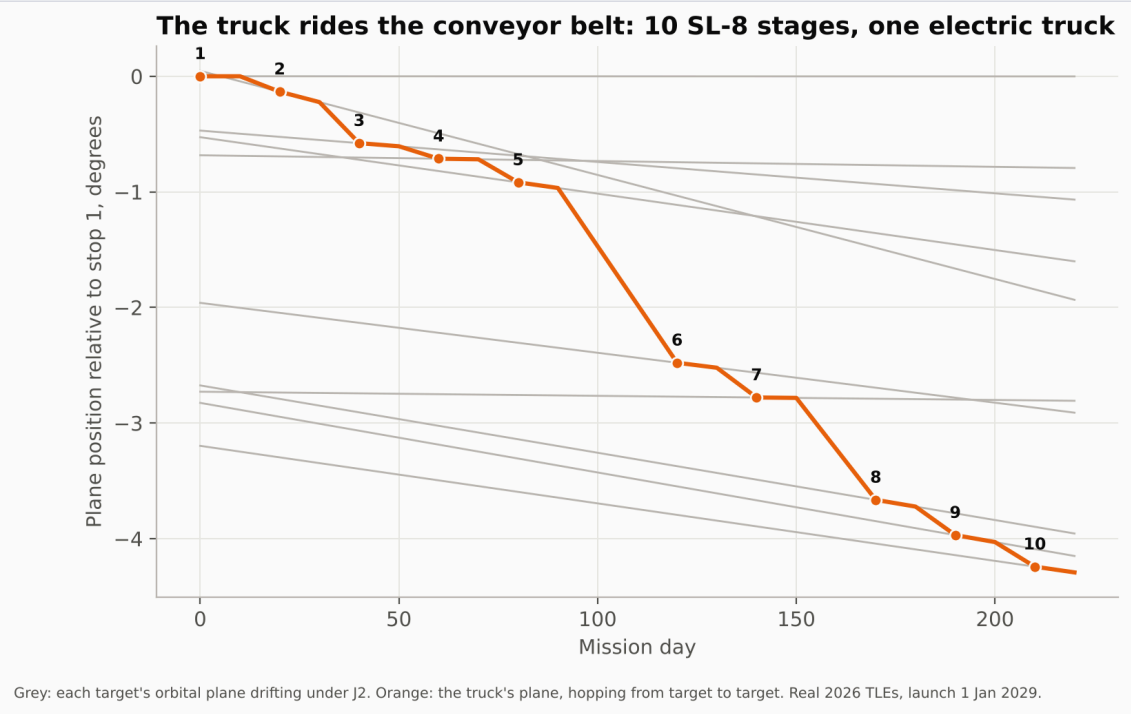


Fig. 5. The optimised electric route through the SL-8 cluster. Grey: each target's orbital plane relative to stop 1; orange: the truck's plane as it hops between targets.

4.3 Validation with SGP4 and numerical integration

Route	Engine	Plane error at meeting, ° (median / max)	Drift-leg correction, m/s (median / max)
SL-8 stages, 83°, ~960 km (Product B)	electric	0.06 / 0.08	–
SL-8 stages, 83°, ~960 km (Product B)	chemical	0.05 / 0.08	2.6 / 10.4
SL-8 stages, 74°, ~750 km	electric	0.89 / 0.97	–
SL-8 stages, 74°, ~750 km	chemical	0.74 / 0.97	60.3 / 86.5
Dead Iridium, 86° (Product A)	electric	0.18 / 0.40	–
Dead Iridium, 86° (Product A)	chemical	0.16 / 0.35	7.9 / 17.0
Partner-nation stages, sun-sync (first route)	electric	0.30 / 0.78	–
Partner-nation stages, sun-sync (first route)	chemical	0.20 / 0.51	7.1 / 46.2

Table 4. Planned rendezvous re-checked with SGP4; drift legs integrated with J_2–J_4 (chemical legs, which use instantaneous burns).

For the ~960 km SL-8 cluster, the target planes predicted by our model agree with SGP4 to 0.06° (median) at the planned meeting times. Correcting the drift legs in a full-gravity integration needs a median of 2.6 m/s, well within the per-stop allowance plus margin. Before we added the SGP4-consistent rate correction, the same check showed errors of about 1°. The lower 750 km SL-8 cluster shows errors over 1° because of atmospheric drag, which our planner does not model. Missions there would need to re-plan each leg with fresh tracking data, as real operators do. The first route agrees to 0.30° (median, worst 0.8°), and its chemical drift legs need a median 7.1 m/s of correction (worst leg 46 m/s). These errors are larger than on the SL-8 route because the truck must hold small tilts for a long time; they are covered by the 10% margin on average but argue for re-planning each leg with fresh tracking data.

4.4 Getting there

Electric truck, 10 stops ($M per removal)	Launch not counted	Rideshare + own climb	Dedicated Neutron	Dedicated Falcon 9
Partner-nation stages, sun-sync (first route)	12.0	**12.2** (80 d, 600 m/s climb; 1,650 kg)	16.4	18.7
SL-8 stages, 83°, ~960 km (Product B)	12.1	**13.0** (375 d, 3,020 m/s climb; 2,374 kg)	16.1	18.5
Dead Iridium, 86° (Product A)	12.0	**12.5** (195 d, 2,327 m/s climb; 1,499 kg)	16.6	18.9

Table 5. Getting to the first stop. Rideshare: SpaceX Transporter-class drop-off at 550 km, 97.6°, $7,500/kg, median over all 24 launch times of day. Dedicated: Neutron $52.5M, Falcon 9 $74M, straight into the first target's orbit. Bold: cheapest.

The rideshare plus the servicer's own climb is the cheapest way to the first stop on every route (Table 5): a dedicated launch adds $3–7M per removal. Its cost depends on how far the first target is from the rideshare orbit. For the first route the climb takes 80 days and 600 m/s. For the SL-8 cluster the servicer must turn its orbit from 97.6° to 83°: 3,020 m/s over 375 days, which raises the launch mass to 2,374 kg and the cost from $12.1M (launch not counted) to $13.0M per removal. A dedicated Neutron would cost $16.1M and a Falcon 9 $18.5M. This is where electric propulsion clearly wins: the chemical truck would launch at 4,083 kg for the same route. The rideshare servicer also exceeds the 831 kg per-port limit, so it needs a custom quote [25].

4.5 Economics

Under base assumptions the first route costs $122M including launch, or **$12.2M per removal** (price $15.3M at a 20% margin). The same cost model, with the same way to orbit, gives $97M for a one-object mission, consistent with

public contracts of $40–93M per object (Fig. 6, Fig. 7). The Monte Carlo puts the cost per removal between $11.2M and $15.4M (P10–P90). The cost per removal falls steeply with the number of stops: $34.0M at 3 stops, $21.8M at 5, $12.2M at 10, $10.0M at 15 and $8.1M at 20. The SL-8 route costs $13.0M and the Iridium route $12.5M, both including launch.

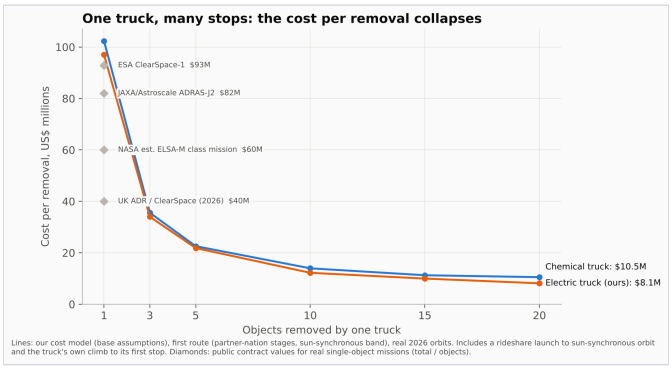


Fig. 6. Cost per removal versus objects removed by one truck (first route, launch included, base assumptions), with public single-object contract values.

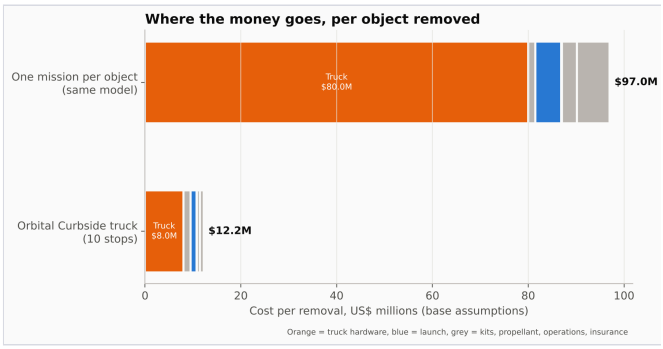


Fig. 7. Where the money goes: fixed costs (mostly the truck) are split ten ways.

Electric versus chemical. Once the route has started, the Pareto frontiers (Fig. 8) show the two engines within a few percent of each other in cost per removal. The electric truck launches about 10% lighter on the SL-8 route itself, but propellant is a small share of the cost. The difference grows when the truck must also change inclination after launch: on the SL-8 route the chemical truck costs $14.2M per removal against $13.0M for the electric one. **Time costs more than fuel:** because operations cost about $5M per year, faster routes are often also cheaper, which is why the planner prices time explicitly.

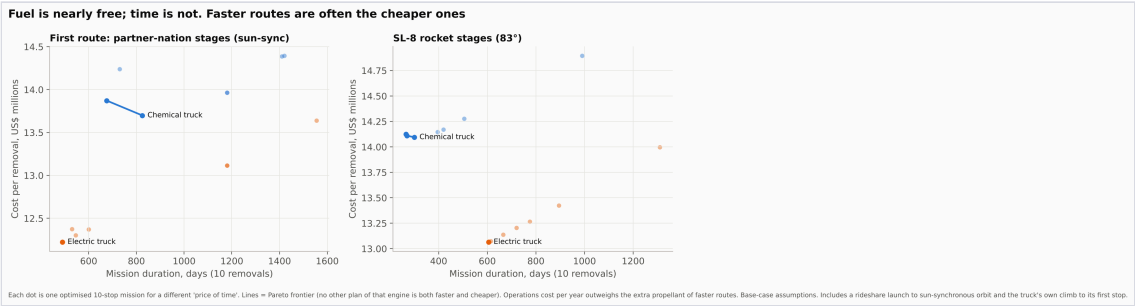


Fig. 8. Mission duration versus cost per removal for many optimised 10-stop missions with different prices of time.

What moves the price (Fig. 9): truck hardware cost ($9.0–16.0M per removal across its range) dominates, followed by kit unit cost, launch price and operations. Xenon price and fuel margin barely matter. **Reuse** (Fig. 10): on the first route, a servicer restocked by kit pods that ride later rideshares can fly 5 routes (50 removals over about 6 years) before its thrusters reach their rated life (5.7 routes), reaching about **$5.2M per removal**. The chemical equivalent costs slightly more ($5.9M at the same number of routes) but needs its propellant delivered and transferred in orbit, against 740 kg of xenon carried from launch by the electric truck. Reuse does not transfer cheaply to the SL-8 cluster, because pods launched on rideshares arrive in the wrong plane.

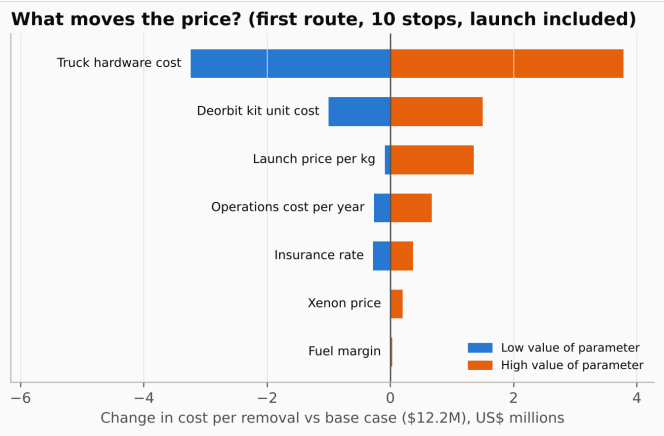


Fig. 9. One-at-a-time sensitivity of cost per removal.

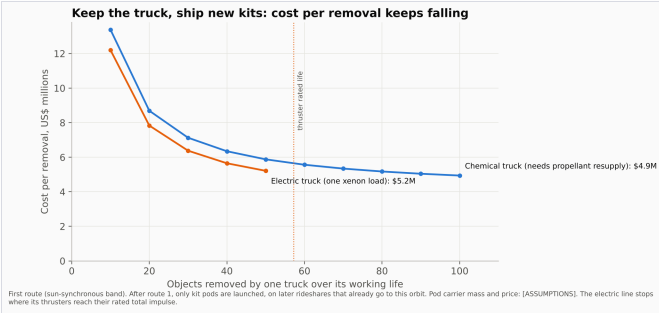


Fig. 10. Cost per removal for one long-lived truck on the first route, restocked with kit pods.

5. DISCUSSION: THE BUSINESS CASE

Two products follow from the analysis. **Product A, Constellation Sweep**, serves operators whose failed satellites cannot meet the five-year rule. Iridium's CEO publicly floated paying $10,000 per deorbit in 2019, while expecting real costs "in the millions or tens of millions, at which price I know it doesn't make sense" [23]. Our model narrows but does not close that gap, so pricing must be validated with operators. **Product B, Rocket-Body Cleanup**, serves agencies that pay to reduce collective risk, the role governments already play in the ESA, JAXA and UK contracts [7–9]. Its first route uses stages owned by the US, India and France in the sun-synchronous band: rideshares go there, reuse works there, and their owners are the same agencies that already buy removal. The SL-8 cluster is the larger prize, but it needs Russia's consent under Article VIII of the Outer Space Treaty and about a year of thrusting to reach. In both cases the value proposition is the same: removal priced like a garbage route, not a one-off moon shot. The strongest structural argument for the electric design is reuse without in-space refuelling.

6. LIMITATIONS

(1) **Dynamics:** circular, secular-only motion. Along-track phasing and proximity operations are an allowance, not simulated, and the planner has no drag (Section 4.3). (2) **Capture:** attaching kits to unprepared, tumbling rocket stages has never been demonstrated; our 10-day stop is an assumption. (3) **Re-entry safety:** uncontrolled re-entry of 1.4 t stages must be shown, object by object, to meet the 1-in-10,000 casualty-risk limit [24]; otherwise heavier controlled-re-entry kits are needed. (4) **Costs:** truck and kit costs are not public and dominate the result; all prices are model estimates, not quotes. (5) **Optimisation:** metaheuristics give good, not guaranteed-optimal, routes, and repeated runs vary by roughly ±10% in duration. (6) **Data:** we used the CelesTrak groups we downloaded, not the full classified catalogue, and our mass data come from public sources. (7) **Launch:** rideshare prices are list prices for up to 831 kg; a 1.5–2.4 t servicer needs a custom quote, and rideshare dates and orbits are set by the launch provider. The first stop is chosen before the launch leg is added; choosing them together could lower the cost further.

7. FUTURE WORK

Interviews with operators, agencies and engineers to replace the assumptions in Appendix A. Collision-risk-weighted target selection (for example using McKnight's ranking [4]). Fleets of trucks sharing kit pods. Drag-aware re-planning for clusters below 800 km. Choosing the first target and the rideshare together. A higher-fidelity low-thrust trajectory optimiser for the final routes.

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APPENDIX A. ASSUMPTIONS

Parameter	Low / Base / High	Source or status
Truck hardware	$50M / $80M / $115M	NASA OTPS 2023 estimates [10]; base assumed
Deorbit kit unit cost	$0.5M / $1.5M / $3M	Assumption, to validate in interviews
Launch price	$7,000 / $7,500 / $15,000 per kg	SpaceX rideshare price list (2026–27); NASA OTPS
Xenon price	$2,780 / $2,780 / $12,000 per kg	Analyst and industry sources
Operations	$3M / $5M / $10M per year	Assumption
Insurance	5% / 8% / 12% of hardware + launch	Assumption
Δv margin	10% / 10% / 25%	From the benchmark and validation results
Truck dry mass; stop time; approach Δv	600 kg; 10 days; 20 m/s	Assumptions
Kit motor	I_sp 285 s; 12 kg + 15% tankage	Assumption (small solid-motor class)
Kit pod carrier (reuse)	150 kg; $4M	Assumption

All code, data and figures are reproducible from the repository (README.md). Results were computed on 16 Oct 2026.